

NURLIANA PARDEDE

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PROFESSIONAL SUMMARY

Chemical Engineering Graduate from Universitas Jambi with a GPA of 3.60/4.00 and experience in refinery operations, materials research, and renewable energy projects. Experienced as a Researcher at Badan Riset dan Inovasi Nasional (BRIN) and Internship Trainee at PT Kilang Pertamina Internasional Refinery Unit III Plaju, with exposure to process evaluation, material characterization, and chemical process analysis. Proficient in process simulation and engineering software including Aspen Plus, Aspen Hysys, ChemCAD, Minitab, and OriginLab for process design, data analysis, and research activities. Seeking an entry-level position in Process Engineering, Production, Quality Control (QC), or Research and Development (R&D).

WORK EXPERIENCE

BADAN RISET DAN INOVASI NASIONAL (BRIN) RI **Nov 2025 – Feb 2026**
Researcher

- Conducted research on hydroxyapatite synthesis from eggshell waste combined with green-synthesized silver nanoparticles (AgNPs) for potential toothpaste applications.
- Prepared hydroxyapatite materials through chemical synthesis processes and optimized calcination parameters to obtain suitable material characteristics.
- Performed material characterization using UV–Vis spectrophotometry, XRD, FTIR, and FESEM to analyze structural, optical, and morphological properties.
- Conducted antibacterial evaluation against *Streptococcus mutans* and *Escherichia coli* using the well diffusion method to assess composite performance.

PT KILANG PERTAMINA INTERNASIONAL REFINERY UNIT III PLAJU **Sep 2025 – Oct 2025**
Internship Trainee

- Studied refinery operations and production processes, including Crude Distillation Unit (CDU), RFCCU, Polypropylene Unit, Polymer Unit, and Alkylation Unit.
- Evaluated Double Pipe Heat Exchanger HE E6-1/2 performance using operational data and assessed material replacement options based on corrosion, thermal efficiency, cost, and reliability aspects.
- Reviewed Process Flow Diagram (PFD) of crude oil refining processes to understand equipment functions and process interconnections.

BACKGROUND EDUCATION

UNIVERSITY OF JAMBI **Aug 2022 – Nov 2026**

Bachelor of Chemical Engineering | (GPA: 3.58/4.00)

- Completed coursework in Mass & Energy Balance, Thermodynamics, Heat & Mass Transfer, Fluid Mechanics, and Process Control & Instrumentation.
- Completed final project titled *"Preliminary Design of a Cyclohexanone Production Plant via Phenol Hydrogenation Process with a Production Capacity of 55,000 Tons/Year."*
- Conducted research on *"Synthesis of Eggshell-Based Hydroxyapatite Doped with Nanosilver (AgNPs) via Green Synthesis Sol-Gel Method for Toothpaste Applications."*

PROJECT EXPERIENCE

Coconut Waste-Based Renewable Energy **May 2024 – Oct 2024**
Student Community Service Program (P2M2), Universitas Jambi
Team Member

- Developed coconut waste-based briquettes as a renewable energy alternative to reduce dependence on conventional fuels.

- Contributed to project implementation, teamwork coordination, and evaluation of renewable energy solutions.

Development of Biomass Biopellets from Water Hyacinth **May 2024 – Oct 2024**
Student Community Service Program (P2M2), Universitas Jambi
Team Member

- Supported the development of water hyacinth biomass-based biopellets as an environmentally friendly alternative energy source.
- Evaluated biomass characteristics and production processes to improve biopellet quality and economic value.

Biopellet Production from Oil Palm Empty Fruit Bunches (EFB) Using Wet Torrefaction **Jun 2024 – Jul 2024**
***Researcher* | Independent Study Program (MBKM), Universitas Jambi**

- Conducted research on converting oil palm empty fruit bunch waste into biomass fuel through wet torrefaction technology.
- Analyzed the effect of operating temperature on moisture content, mass yield, and fuel characteristics.

ORGANIZATIONAL EXPERIENCE

HIMPUNAN MAHASISWA TEKNIK KIMIA (HIMATEKA) UNIVERSITAS JAMBI **Dec 2022 – Aug 2024**
Staff of Organizational Development & Cadre Development

- Assisted in planning and executing organizational work programs, including coordination, event preparation, and activity evaluation.
- Developed leadership, communication, time management, and problem-solving skills through active involvement in organizational activities.

Staff of Secretariat Department ***Sep 2024 – Jul 2025***

- Managed organizational administration, including official correspondence, document filing, and record management.
- Organized meeting documents, activity reports, and organizational inventories to support operational efficiency.

CHEMICAL ENGINEERING FAIR (CEF) 2023 & 2024
Staff of Public Relations Division **Oct 2022 – May 2023**

- Prepared sponsorship proposals and communication materials for external partners and potential sponsors.
- Maintained communication and follow-up with sponsors to support event collaboration.

Treasurer – Scientific Writing & Essay Competition Committee ***Sep 2023 – May 2024***

- Managed event financial administration, including budget planning, transaction records, and financial documentation.
- Prepared financial reports to ensure transparent and accountable event management.

CERTIFICATIONS & TRAINING

- **Process Safety Engineering Course | Ikatan Alumni Teknik Kimia (IATEKA), Universitas Jambi, 2026**
  Covered process safety concepts including PFD & P&ID Reading, Loss Prevention, Fire & Explosion Engineering, HAZID, and HAZOP Risk Assessment to support the implementation of safety practices in chemical process industries.
- **AutoCAD Plant 3D for Plant Design Training | Ikatan Alumni Teknik Kimia (IATEKA), Universitas Jambi, 2024**
  Learned 3D plant design, equipment modeling, and piping layout development to support process plant design and engineering applications

SKILLS

- **Process Engineering:** Process Simulation, Process Design, Process Flow Diagram (PFD) Analysis, P&ID Reading, Heat Exchanger Evaluation, Process Optimization
- **Engineering Software:** Aspen Plus, Aspen HYSYS, ChemCAD, SuperPro Designer, MATLAB, AutoCAD Plant 3D, Minitab, OriginLab
- **Laboratory & Material Analysis:** XRD, FTIR, FESEM, UV–Vis Spectrophotometry, Material Characterization, Experimental Analysis
- **Data & Productivity Tools:** Microsoft Excel, Word, PowerPoint, Data Processing & Visualization
- **Professional Skills:** Analytical Thinking, Problem Solving, Team Collaboration, Technical Communication, Time Management
- **Languages:** Indonesian (Native), English (Upper-Intermediate, TOEFL Prediction Score: 550)